

GRPO

$$A_{\text{GRPO}}^i = \frac{R^i - \bar{R}}{\sqrt{\widehat{\text{Var}}(R)} + \varepsilon} = \frac{\sum_{l=1}^r (R_l^i - \bar{R}_l)}{\sqrt{\widehat{\text{Var}}\left(\sum_{l=1}^r R_l\right)} + \varepsilon} = \frac{\sum_{l=1}^r (R_l^i - \bar{R}_l)}{\sqrt{\sum_{l=1}^r \sum_{m=1}^r \widehat{\text{Cov}}(R_l, R_m)} + \varepsilon}.$$

CorrGRPO

$$A_{\text{CorrGRPO}}^i = \frac{\sum_{l=1}^r (R_l^i - \bar{R}_l)}{\sqrt{\sum_{l=1}^r \sum_{m=1}^r \hat{\rho}_{lm}} + \varepsilon} = \frac{\sum_{l=1}^r (R_l^i - \bar{R}_l)}{\sqrt{\sum_{l=1}^r \sum_{m=1}^r \frac{\widehat{\text{Cov}}(R_l, R_m)}{\sqrt{\widehat{\text{Var}}(R_l)} \sqrt{\widehat{\text{Var}}(R_m)}}} + \varepsilon}.$$

Notation

- n : number of rollouts in the group; $i = 1, \dots, n$ indexes rollouts.
- r : number of reward components; $l, m = 1, \dots, r$ index reward components.
- R_l^i : reward component l of rollout i ; $\bar{R}_l = \frac{1}{n} \sum_{k=1}^n R_l^k$ is its group mean.
- $R^i = \sum_{l=1}^r R_l^i$: total reward of rollout i ; $\bar{R} = \frac{1}{n} \sum_{k=1}^n R^k$ is its group mean.
- $\widehat{\text{Cov}}$, $\widehat{\text{Var}}$, and $\hat{\rho}_{lm}$: sample covariance, variance, and Pearson correlation coefficient, computed across the group.
- $\varepsilon > 0$: numerical stability constant.

